

MOHAMMED MUBARAK

Chennai, Tamil Nadu, India | +91 7569319827 | mubarak7842a@gmail.com

GitHub: github.com/mubarak6969 | LinkedIn: linkedin.com/in/mohammed-mubarak-0522a1257

PROFESSIONAL SUMMARY

B.Tech Computer Science graduate (2025, CGPA 8.9/10) with strong fundamentals in Python, data structures and algorithms, SQL databases, and backend development, including REST APIs and HTTP. Completed internships in Web Development and Machine Learning, and built projects that add practical AI/ML skills such as machine learning algorithms and computer vision. Seeking an entry-level Backend / Python Developer role to build reliable software and keep learning from experienced engineers.

TECHNICAL SKILLS

Frontend: HTML, CSS, JavaScript, TypeScript

Backend: Python, Java, FastAPI, PostgreSQL

Backend Core Concepts: REST API, HTTP Status Codes, Deployment, CRUD Operations

Databases: SQL, MySQL

Data & ML: NumPy, Pandas, Matplotlib, Seaborn, OpenCV, Machine Learning Algorithms

Version Control: Git, GitHub

PROJECTS

ECG Intelligence: ECG Classification using Deep Learning | 2025 | [GitHub](#)

Python, TensorFlow/Keras, REST API

- Trained a Conv1D-LSTM model to classify ECG signals from a 21,799-record dataset, reaching 80.1% accuracy and 0.889 ROC-AUC on unseen patient data.
- Found and corrected a labeling error affecting about 40% of the records before training.
- Wrapped the model in a REST API, containerized it with Docker, and wrote 63 pytest tests. Research paper published in IJSART, 2025.

Deep-Dive Knowledge Assistant: Question Answering over Videos, PDFs and Audio | 2025 | [GitHub](#)

Python, Sentence-Transformers, ChromaDB

- Built a question-answering tool that converts audio to text with Whisper, stores content embeddings in ChromaDB, and retrieves relevant passages for each question.
- Combined keyword search (BM25) with embedding search to improve the relevance of retrieved results.
- Answers include source references, and the tool says "insufficient evidence" when nothing relevant is found.

Facial Recognition Attendance System | 2025 | [GitHub](#)

Python, OpenCV, MediaPipe, PostgreSQL, REST API

- Built a web app that marks attendance by recognizing faces from a webcam, with separate login roles for admin and users.
- Fixed a bug where simultaneous requests created duplicate attendance entries; wrote 66 pytest tests.
- Deployed online using Render and Neon PostgreSQL.

INTERNSHIP EXPERIENCE

Machine Learning Intern - VCodez - Innovating Ideas | Chennai, India | Jun 2025 - Sep 2025

- Built a facial recognition attendance system in Python using OpenCV and MediaPipe, with a web interface and a PostgreSQL database.
- Cleaned data and created features with Pandas and NumPy to prepare datasets for model training and evaluation.
- Wrote SQL queries to extract and prepare data for model testing, and tested the application with 66 pytest tests.

Web Development Intern - Motion Cut | Remote | Nov 2023 - Dec 2023

- Tested 15+ REST API endpoints using Postman to check that responses and status codes were correct.
- Worked with MySQL tables and built responsive pages using HTML, CSS, and JavaScript.

EDUCATION

B.Tech, Computer Science Engineering - Bharath Institute of Science and Technology, Chennai | 2021 - 2025
CGPA: 8.9 / 10

CERTIFICATIONS

- Data Analytics Job Simulation, Deloitte (Forage), 2025
- The Complete MySQL Bootcamp, Udemy, 2024
- Python for Beginners, Udemy, 2024

ACHIEVEMENTS

- Published research paper "ECG Peak Detection Using Deep Learning Approaches" in IJSART, Vol. 11, Issue 4, 2025 (Paper ID: IJSARTV11I4103081).
- Solved 100+ LeetCode problems covering arrays, strings, hashing, searching, sorting, and optimization.